

Module 4

Engineering Materials

What makes chemistry important in engineering?

“Chemistry helps us to make and understand the properties of materials”

Materials offer a wide range of properties, for example;

1. Electrical and thermal conductivities
2. Mechanical properties
3. Magnetic properties
4. Optical properties (light absorption and emission properties)

Importance of Materials

Bike-Arantix (made by Delta 7)



First lightweight mountain bike

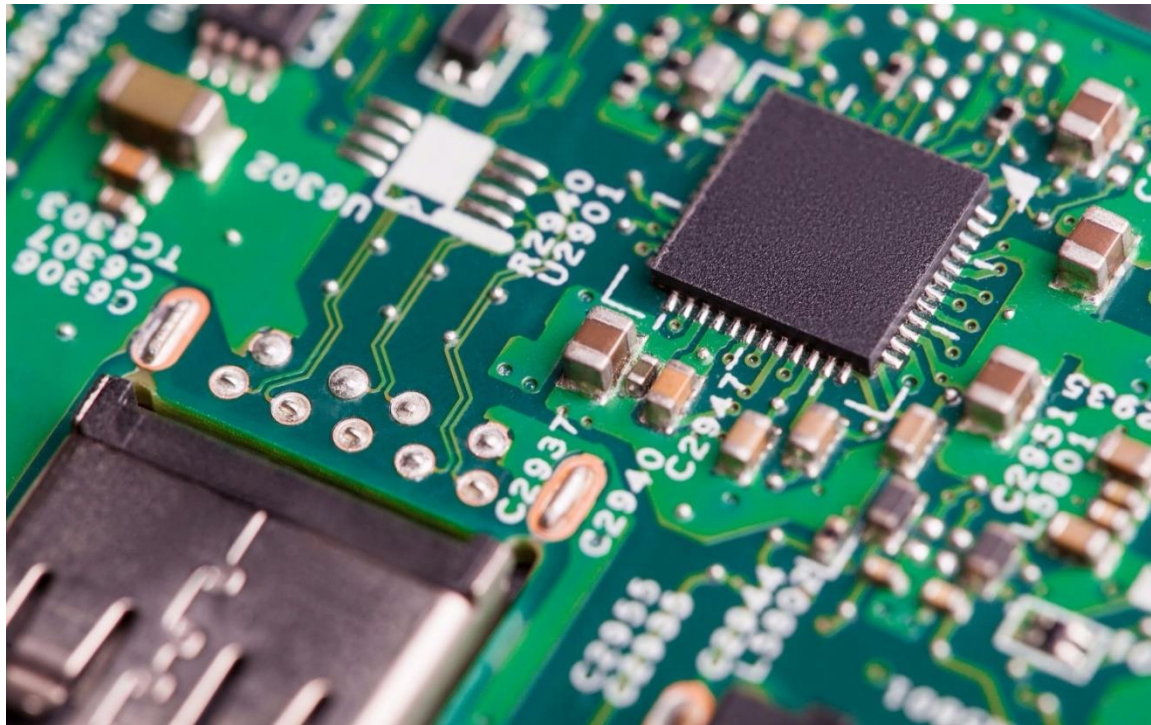
2007

Salient Features:

1. IsoTuss Carbon Fiber
2. Kelval spider web-like open lattice tube design.
3. Handcrafted in US, weighs only 2.74 pounds.

Importance of Materials

A variety of materials are used to build a functioning circuit board. That includes, **insulators, conductors and **semiconductors**.**



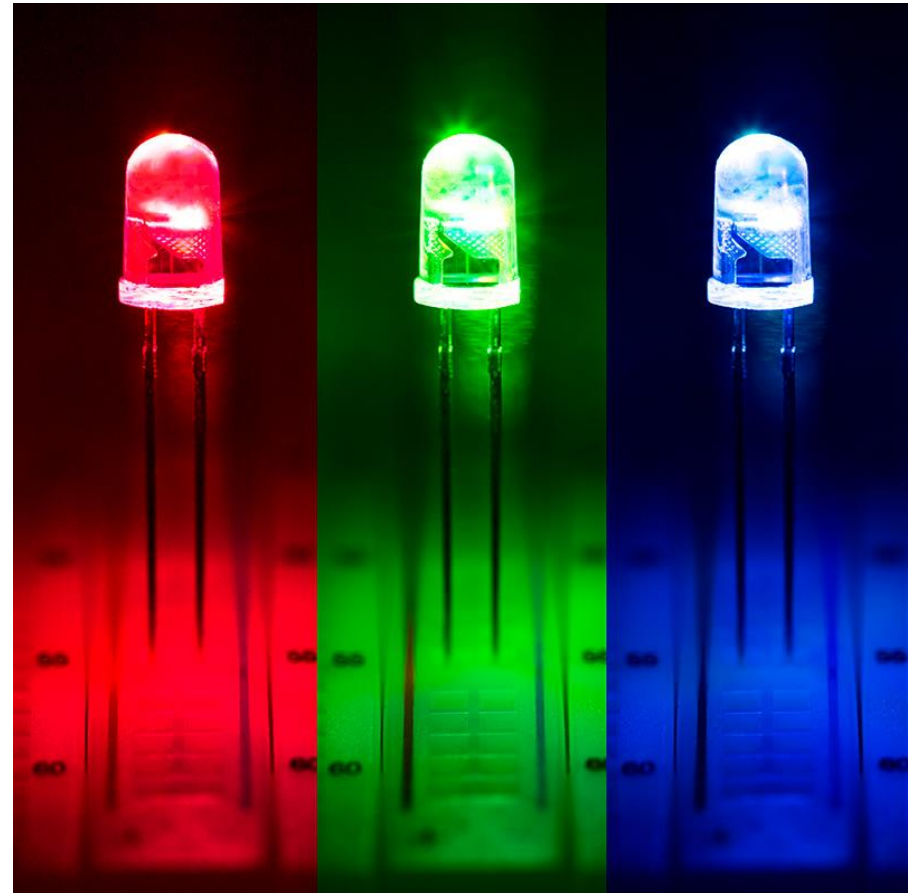
Resistors, *diodes*, and *transistors* are integral parts of a functioning *Printed Circuit Board*.

Importance of Materials

Light emitting diode is a solid state semiconductor device that converts **electrical energy** into **light energy**.

LEDs are used in many object that we use everyday.

From television remote control to traffic lights



What makes a solid to be a **conductor** or **semiconductor** or an **insulator**?

As we shall see, the *type of bonding in a solid* is directly related to the conductivity of the solid.

Metals, **non-metals**, and **semimetals** have different bonding properties that lead to the differences in conductivity that can be observed between these categories of elements.

Therefore, the concepts of **chemical bonding** is crucial to tune the properties of the materials used in engineering designs.

Covalent bonding

Ionic Bonding

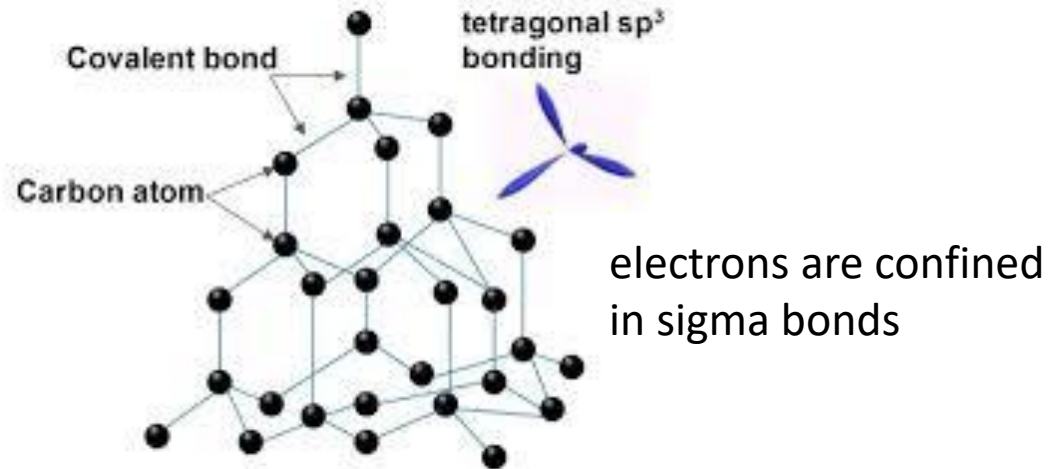
Metallic Bonding

Covalent bonding

- Electrons are **shared** between pairs of atoms



Diamond



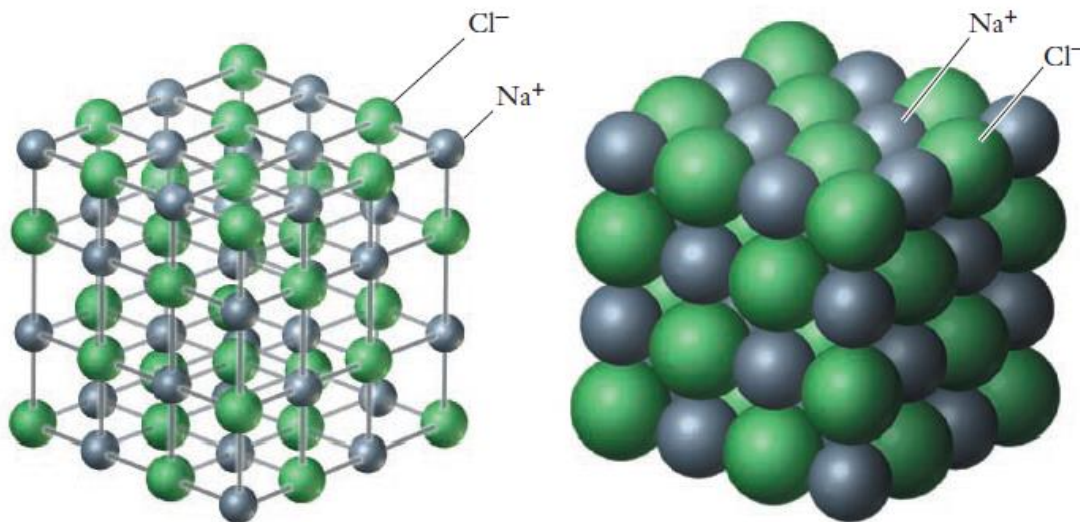
When the atoms are bound by strong covalent bonds, the valence electrons are localized.
consequently they will **not conduct electricity**.

Ionic Bonding

Valence shell electrons are confined on ions, **electrostatic force** holds the ions together.

Example **NaCl**

Two equivalent representations of NaCl

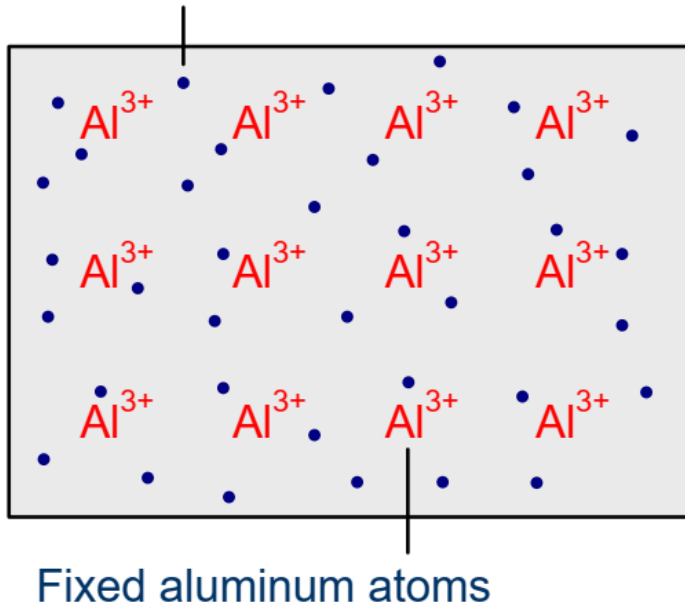


Ionic compounds form extended systems or lattices of alternating positive and negative charges. They *do not have electronic conductivity*.

Metallic Bonding

Metals represent another type of extended lattice system, but here the chemical bonding is totally different.

Free electrons form an electron gas



The nuclei and some fraction of their electrons comprise a positively charged “**core**” localized at these lattice points, and other electrons move more or less freely throughout the whole array.

This is called **metallic bonding**.

This model is called **Electron-Sea Model**

Electrons are not localized or confined.

These free electrons (*valence electrons*) are responsible for high electrical conductivity of the metals.

Electrical Conductivity: **Metals**

All metals can be drawn into wires (*ductility*) or beaten into sheets (*malleability*)

All metals have a high *thermal* and *electrical conductivity*

The s-block metals occur as chlorides, silicates, and carbonates.

1A	2A											3B	4A	5A	6A	7A	8A
													Si	P			
Li	Be											Al					
Na	Mg	3B	4B	5B	6B	7B	8B		1B	2B		Ga					
K	Ca	Sc	Ti	V	Cr	Mn	Fe	Co	Ni	Cu	Zn						
Rb	Sr	Y	Zr	Nb	Mo	Tc	Ru	Rh	Pd	Ag	Cd	In	Sn				
Cs	Ba	Lu	Hf	Ta	W	Re	Os	Ir	Pt	Au	Hg	Tl	Pb	Bi			

The *d*- and *p*-block metals are found as oxides and sulfides, except for the group 3B metals, which occur as phosphates, and the platinum-group metals and gold, which occur in uncombined form.

There is no mineral source of technetium, a radioactive element that is made in nuclear reactors.

	
Chlorides	Silicates
	
Carbonates	Phosphates
	
Oxides	Free metals
	
Sulfides	

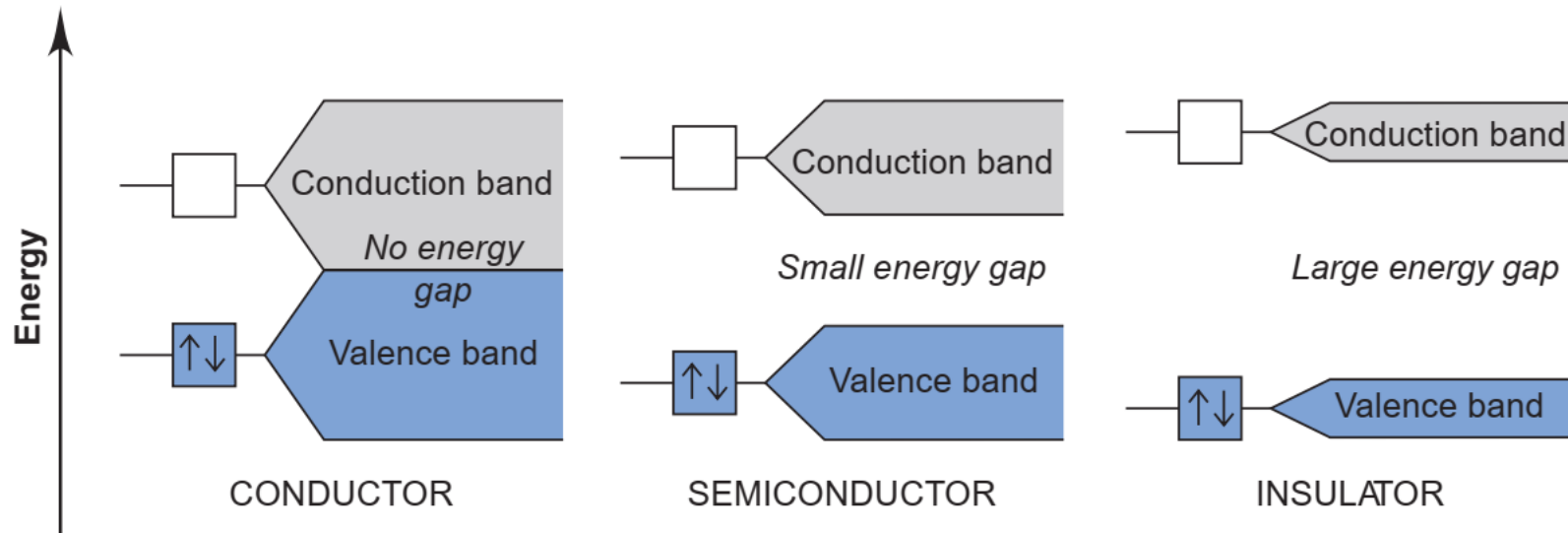
Classifications of materials-Based on Electrical Conductivity

Based on electrical conductivity we can classify the materials into three types.

- 1) Conductors (Metals and graphite)
- 2) Semiconductors (Silicon, GaAs, InP, etc.)
- 3) Insulators (Ceramic oxides, Polymers, Paper)

Band Gapes in Different Materials

The energy band structure of a solid determines whether it is a conductor, an insulator or a semiconductor.



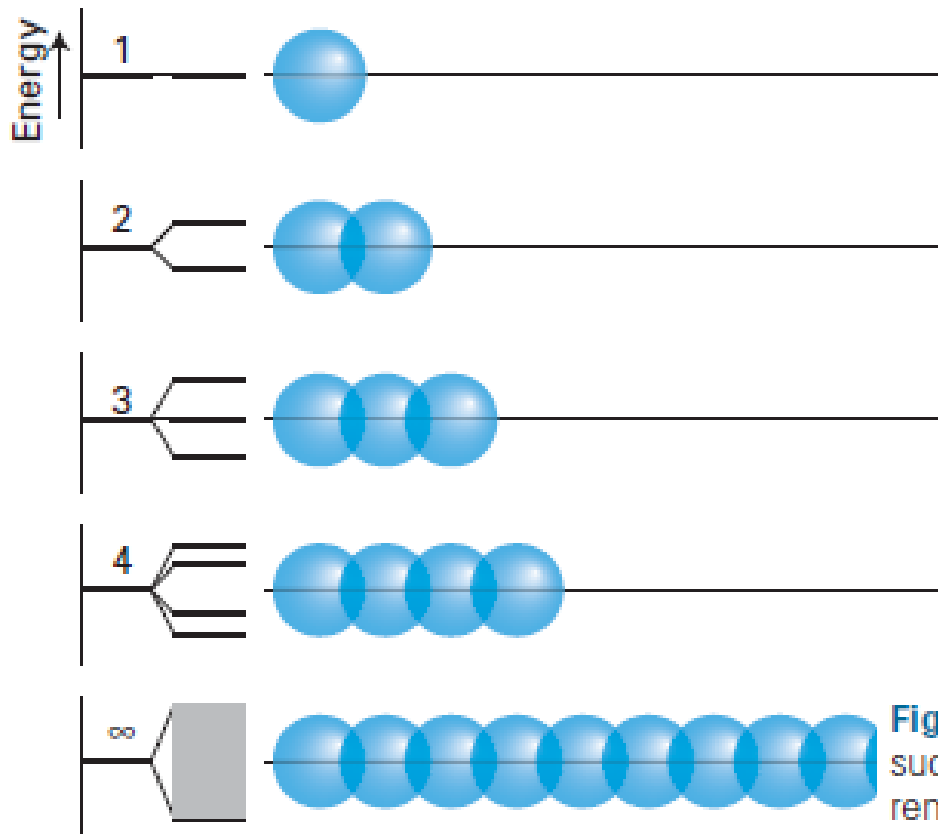
1. What is band?

2. How the valence and conduction band forms?"

Band Theory of Solids

A bands is a group of very closely spaced orbitals with not having much difference in energy.

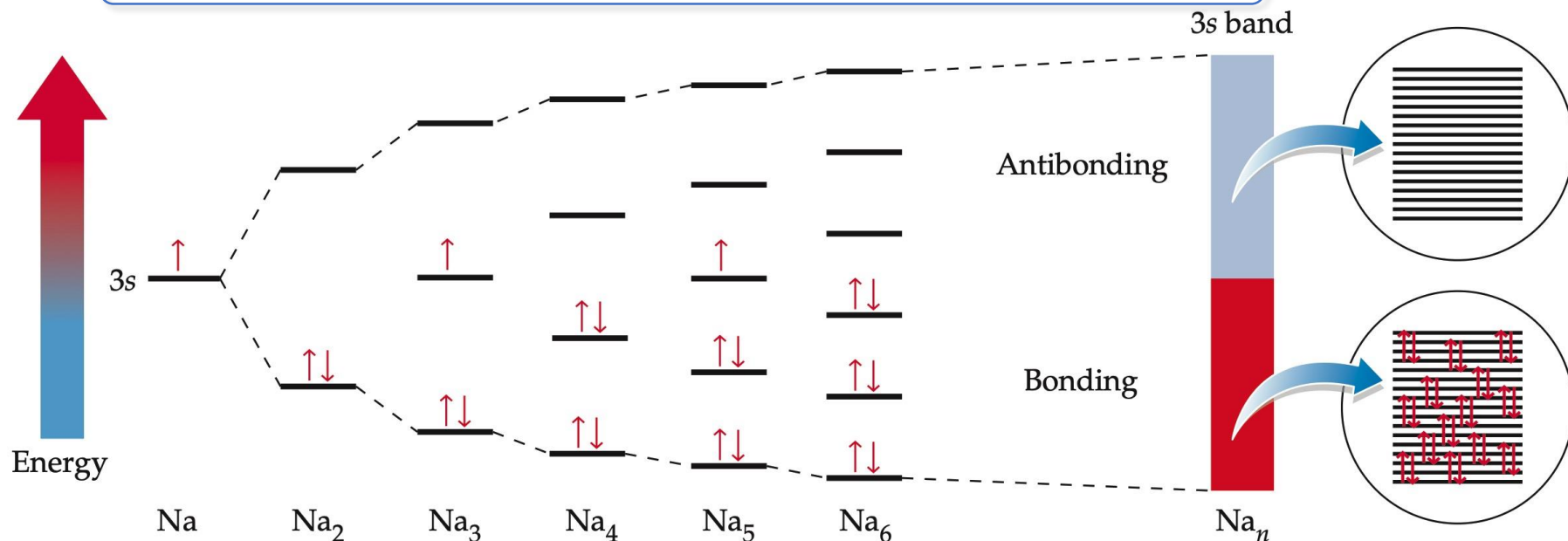
They are formed by the successive addition of **atomic orbitals** of multiple atoms.



**N-atomic orbitals combine
to form N-Molecular orbitals**

Fig. 17.2 The formation of a band of N molecular orbitals by successive addition of N atoms to a line. Note that the band remains of finite width, and although it looks continuous when N is large, it consists of N different orbitals.

As the value of n increases, the energy levels merge into an almost continuous band.

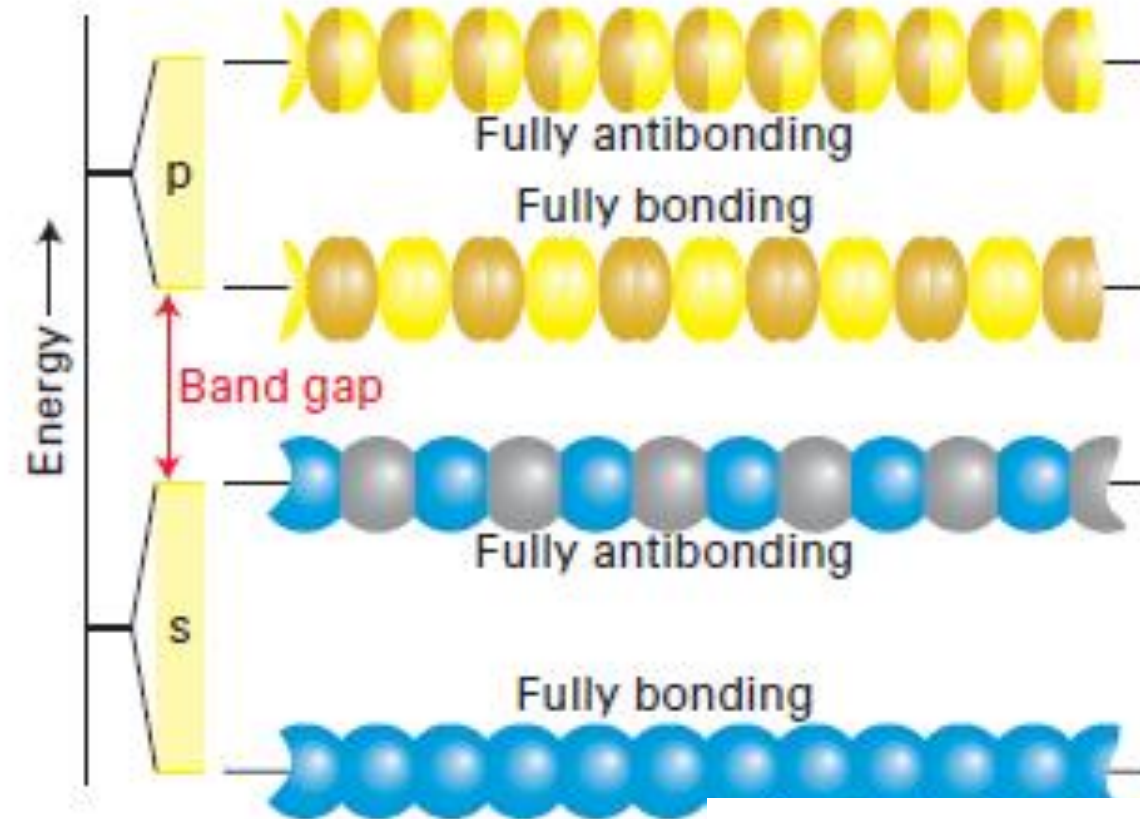


The difference in energy between successive MOs in an Na_n molecule **decreases** as the number of **Na atoms increases**

The **lower-energy half** of the 3s band for Na consists of closely spaced bonding MOs, which are **filled**

The **higher-energy half** of the 3s band for Na consists of closely spaced antibonding MOs, which are **empty**

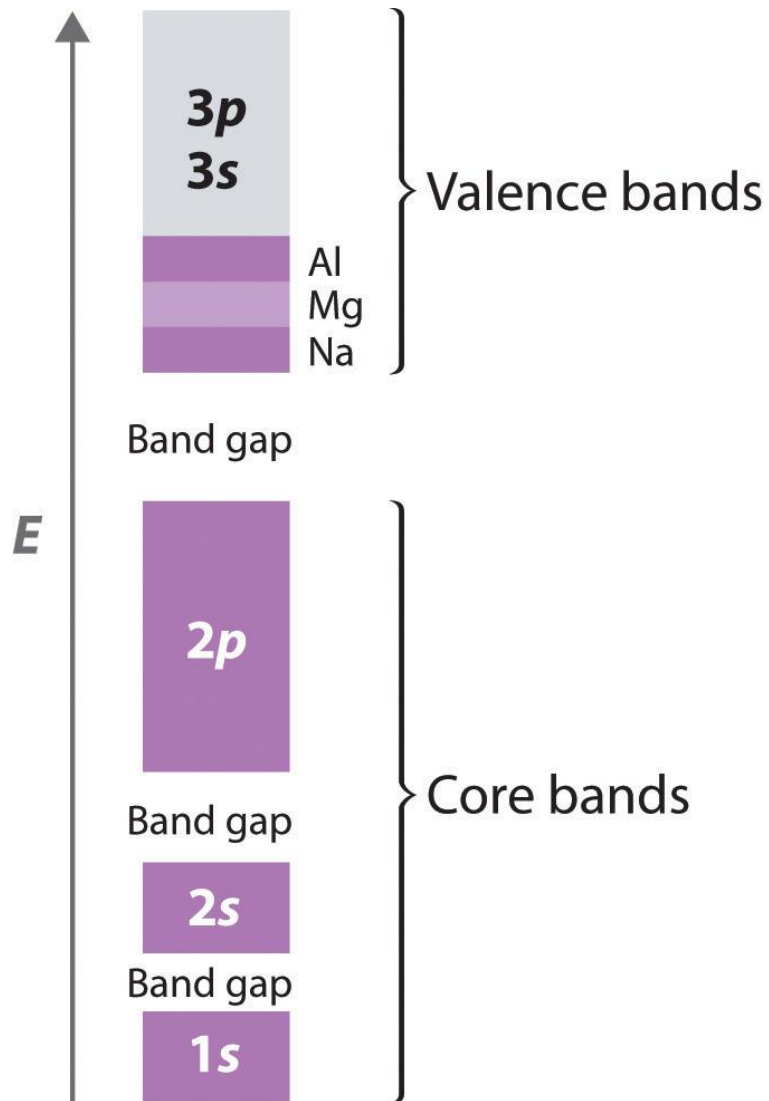
s Band and p Band



The overlap of '**s orbitals**' gives to an '**s band**'

Fig. 17.3 The overlap of s orbitals gives rise to an s band, and the overlap of p-orbitals gives rise to a p band. In this case the s and p orbitals of the atoms are so widely spaced that there is a band gap. In many cases the separation is less, and the bands overlap.

Overlap of 3s and 3p orbitals in Na, Mg, and Al



Condition for electrical conductivity

“The valence or conduction band or both must be partially filled”

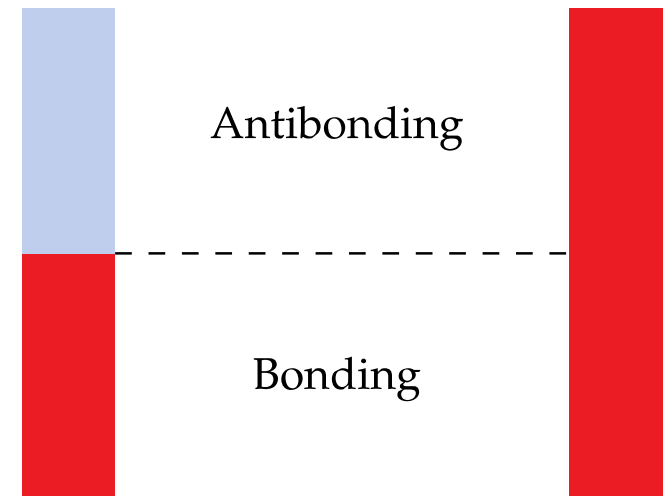
- For metals the valence and conduction bands are overlapped (*filled and empty band orbitals are overlapped*)
- If **band gap** is very **high**, then the material will be an **insulator**.

An electrical potential can shift electrons from one set of energy levels to the other **only if the band is partially filled**

If the band is **completely filled**, there are **no available vacant energy levels to which electrons can be excited**, and therefore the two sets of levels must remain equally populated, even in the presence of an electrical potential

Materials that have only **completely filled bands** are therefore **electrical insulators**.

By contrast, materials that have **partially filled bands** are **metallic conductors**



How to increase conductivity of a semiconductor?

- **Increase temperature**
- **Doping**

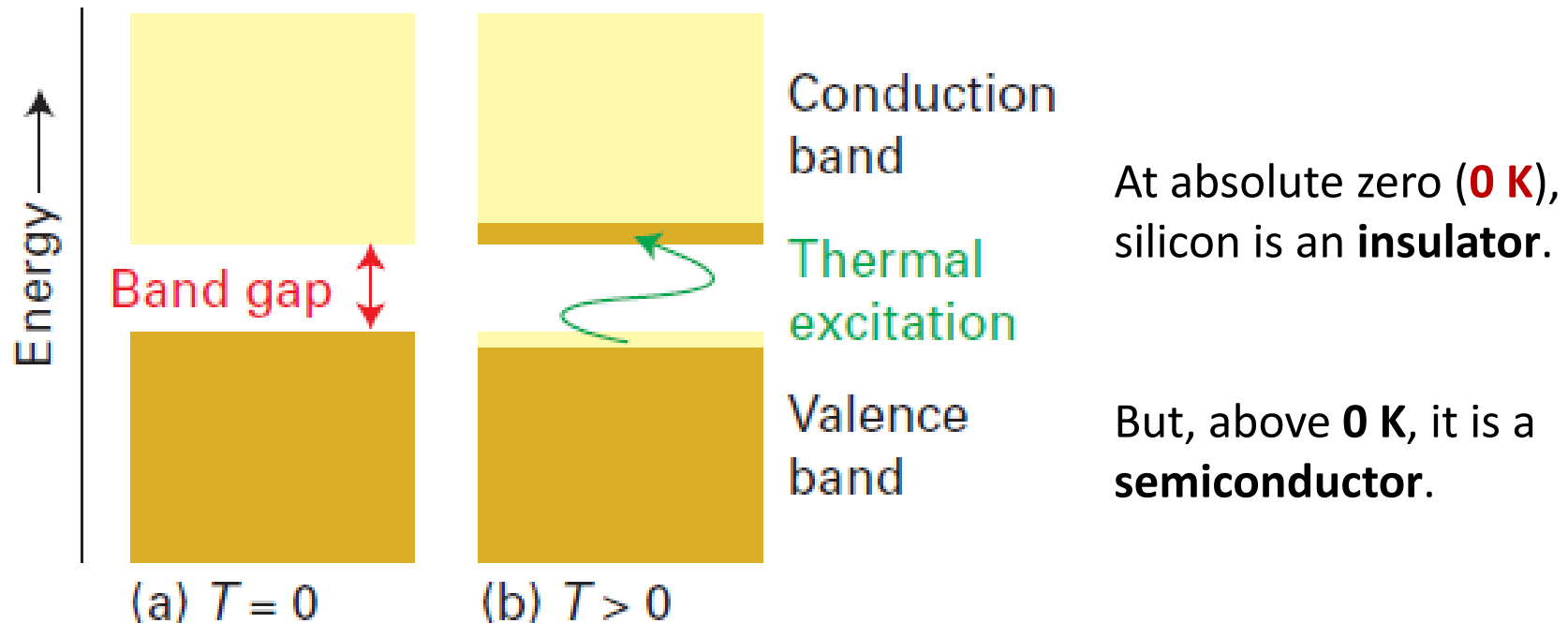


Fig. 17.5 (a) When $2N$ electrons are present, the band is full and the material is an insulator at $T = 0$. (b) At temperatures above $T = 0$, electrons populate the levels of the conduction band at the expense of the valence band, and the solid is a semiconductor.

Semiconductor Types

Semiconductors can be classified as *intrinsic* or *extrinsic* semiconductors.

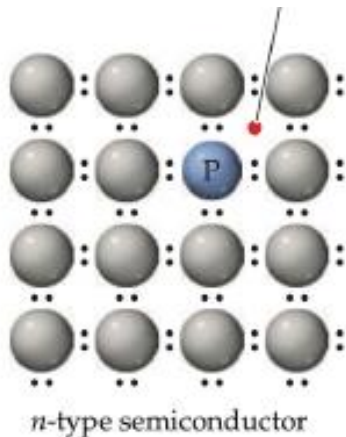
- An **intrinsic** (pure) semiconductor, also called an un-doped semiconductor or **i-type semiconductor**.

Extrinsic Semiconductors

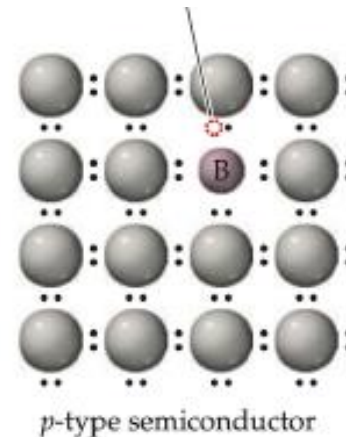
- *Extrinsic semiconductors* are obtained by doping of intrinsic semiconductor with small amount of impurities called *dopant*.

Such, doping enhances the **conductivity** of a intrinsic semiconductor.

Doping of silicon with a group 15 (or 5A) element gives a *n-Type semiconductor*



Doping of silicon with a group 13 (or 3A) element gives a *p-Type semiconductors*

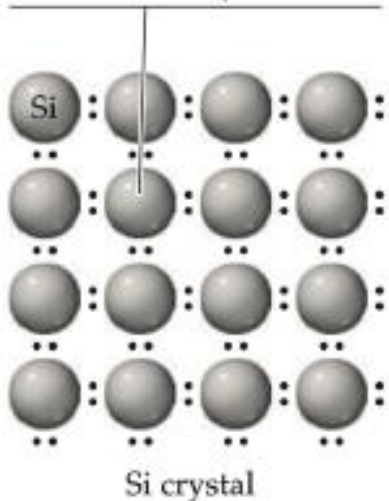


Extrinsic Semiconductors

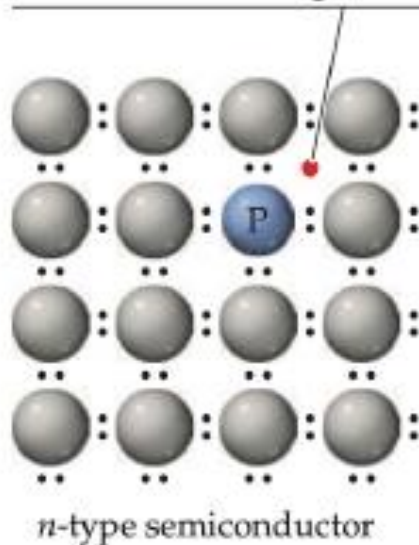
Doping of silicon with a group 15 (or 5A) element gives a *n-Type semiconductor*

Doping of silicon with a group 13 (or 3A) element gives a *p-Type semiconductors*

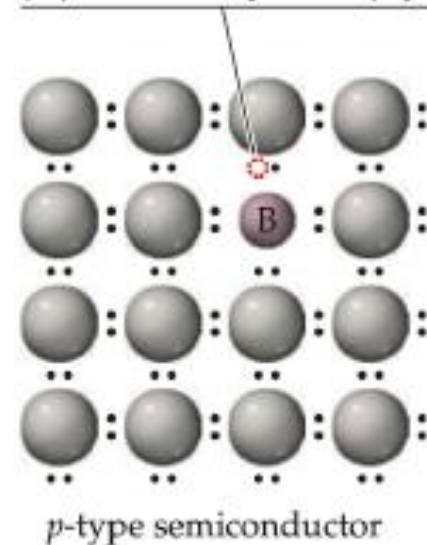
Silicon forms bonds to four other silicon atoms and has a complete octet.



A silicon crystal doped with a P atom has an extra electron that is not needed for bonding. The *negative* electron is the charge carrier.



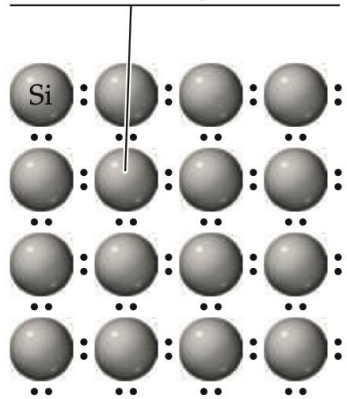
A silicon crystal doped with a B atom is electron deficient. The charge carrier is called a *positive* hole (represented by an empty circle.)



$$3s^1 3(PxPyPz)^3$$

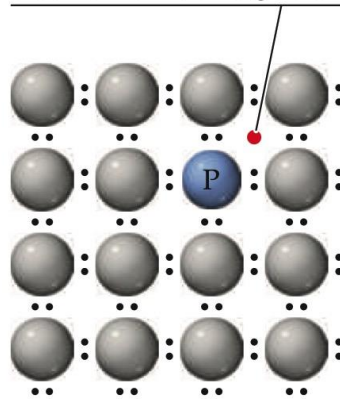
How energy bands changes upon doping?

Silicon forms bonds to four other silicon atoms and has a complete octet.



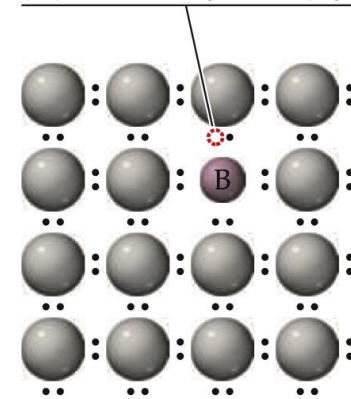
Si crystal

A silicon crystal doped with a P atom has an extra electron that is not needed for bonding. The *negative* electron is the charge carrier.



n-type semiconductor

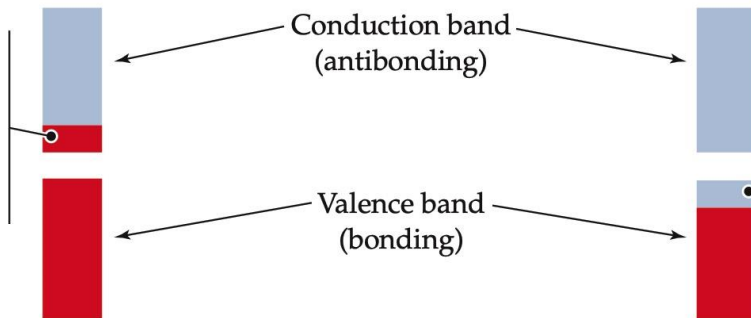
A silicon crystal doped with a B atom is electron deficient. The charge carrier is called a *positive* hole (represented by an empty circle.)



p-type semiconductor



An *n*-type semiconductor, such as silicon doped with phosphorus, has more electrons than needed for bonding and thus has more electrons in the partially filled conduction band than pure silicon.

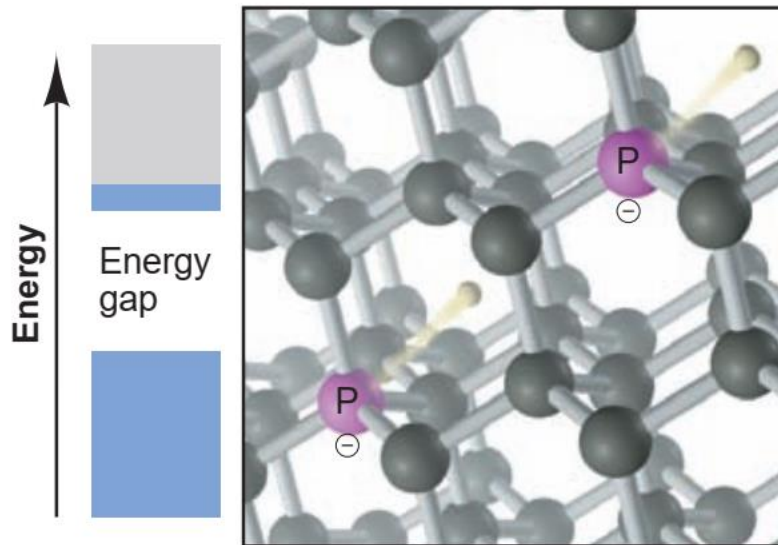


Conduction band
(antibonding)

Valence band
(bonding)

A *p*-type semiconductor, such as silicon doped with boron, has fewer electrons than needed for bonding and thus has more vacancies—positive holes—in the valence band than pure silicon.

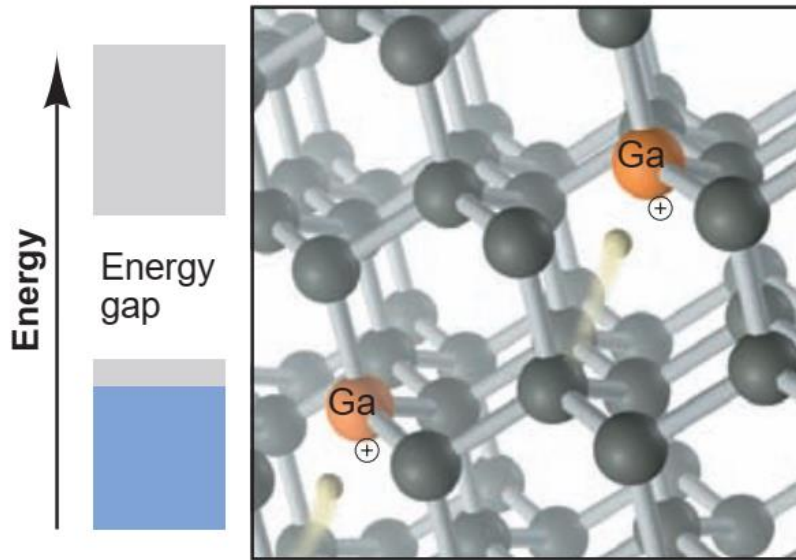
Doping: n-type semiconductor



B n-Type doping with phosphorus

- When **Si** is doped with **phosphorus** [or another Group 5A(15) element], **P** atoms occupy some of the lattice sites.
- Since **P** has **one more** valence electron than **Si**, this additional electron must enter an empty orbital in the **conduction band**, thus bridging the energy gap and **increasing conductivity**.

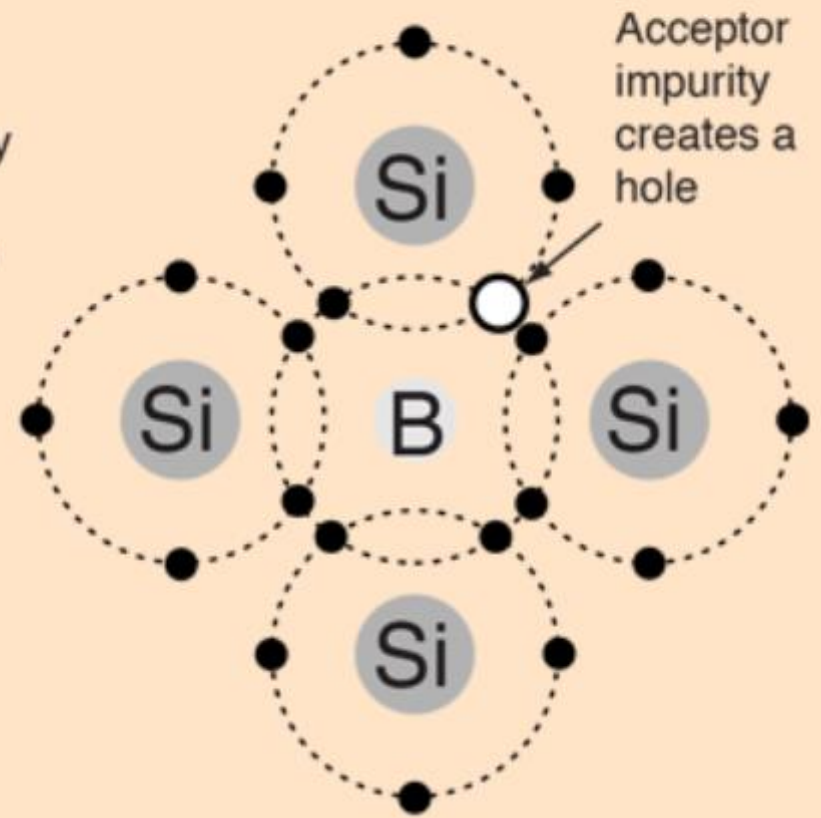
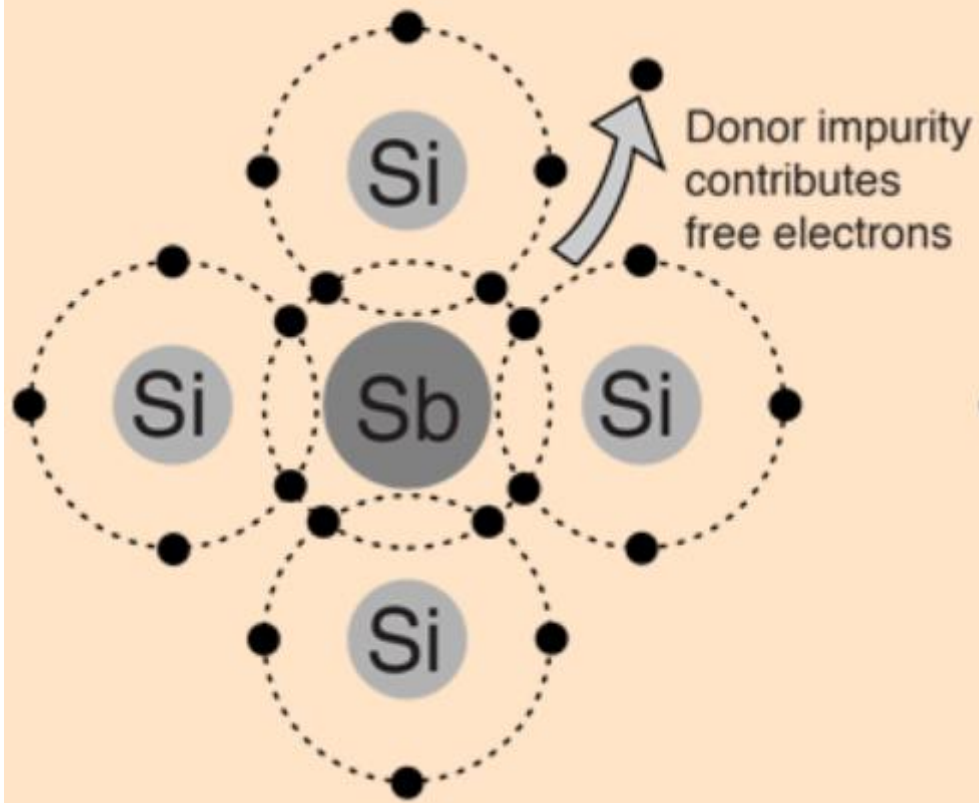
Doping: p-type semiconductor



C p-Type doping with gallium

- When **Si** is doped with **gallium** [or another Group 3A(13) element], **Ga** atoms occupy some sites.
- Since **Ga** has one fewer valence electron than **Si**, some of the orbitals in the valence band are empty, which creates a positive site.

P- and N- Type Semiconductors



Temperature dependence of electronic conductors

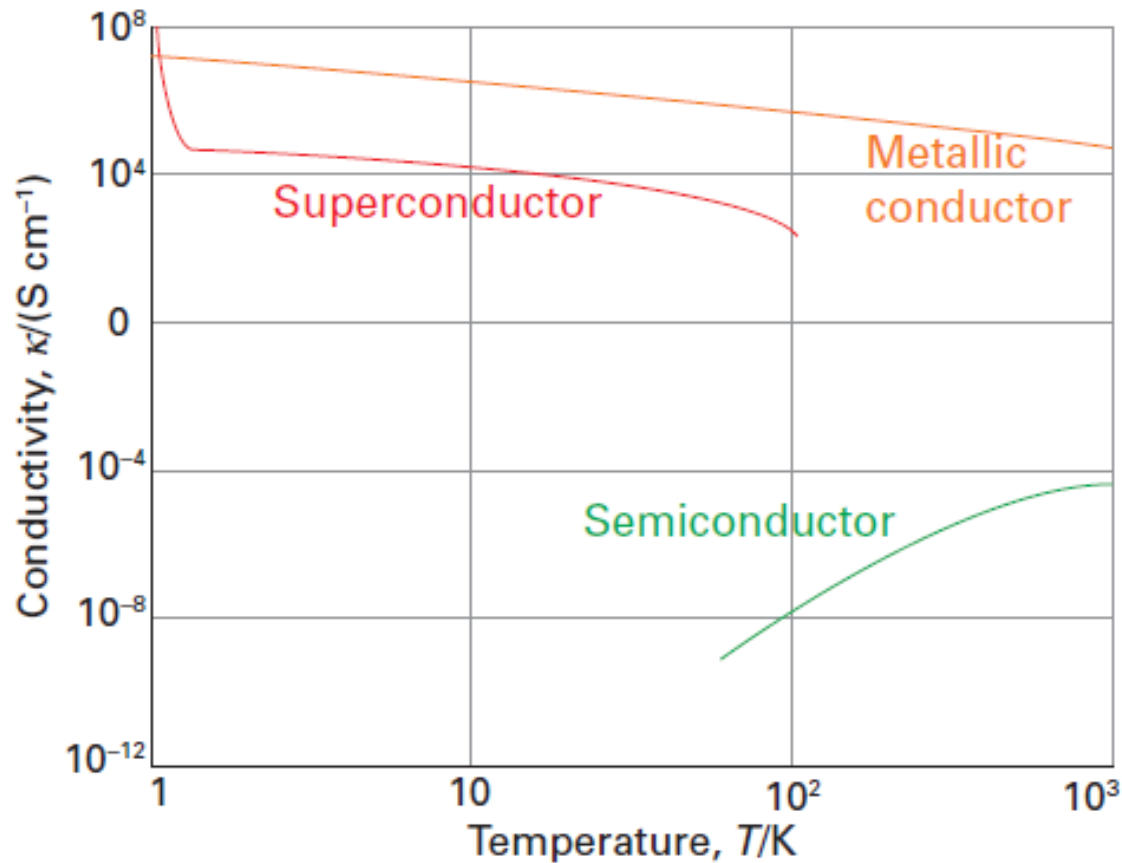
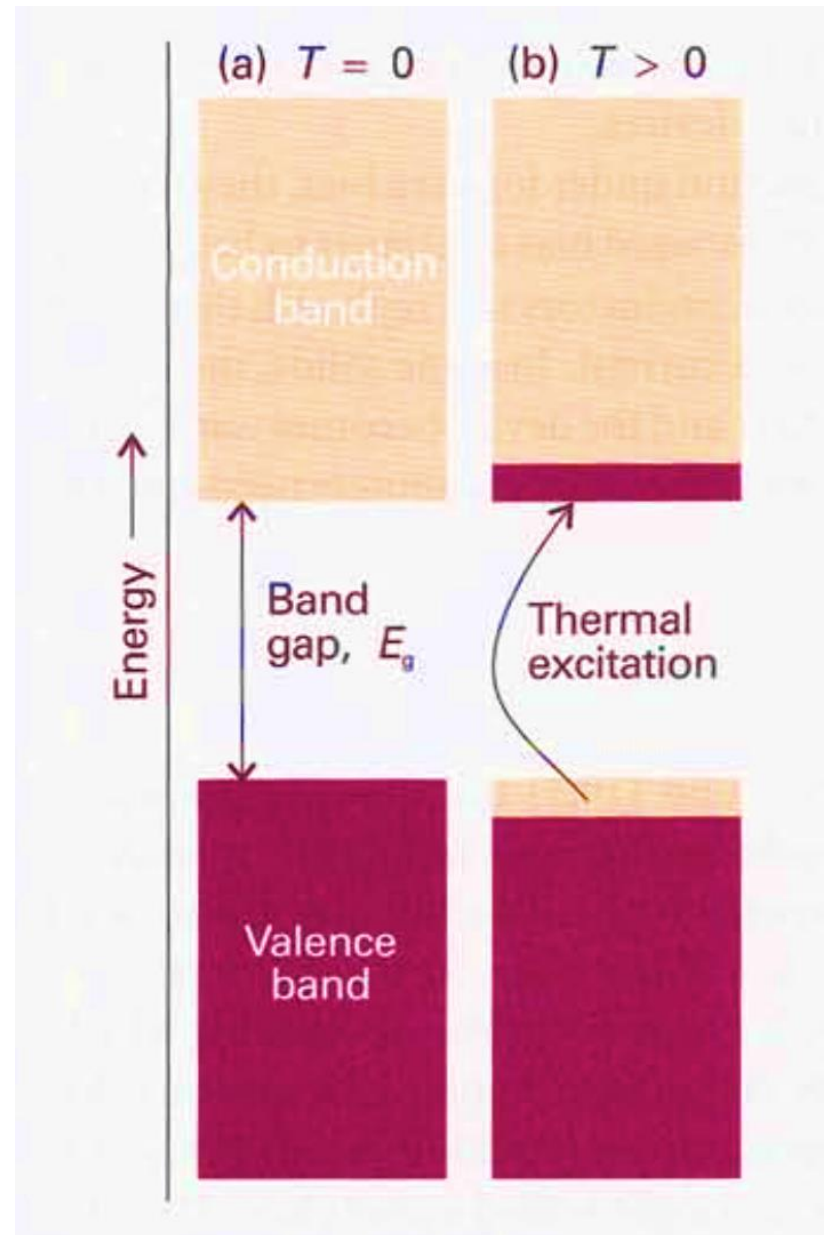


Fig. 17.1 The typical variation with temperature of the electrical conductivities of different classes of electronic conductor.



Why conductivity of **semiconductor** increases with **T** and conductivity of **metals** decreases with **T**?

The electrical conductivity of a semiconductor **increases** with increasing temperature:

The **number of electrons** with sufficient thermal energy to occupy the conduction band increases as the temperature rises

The electrical conductivity of a metal **decreases** with increasing temperature:

At higher temperatures, the metal cations undergo increased vibrational motion about their lattice sites and vibration of the cations **disrupts the flow** of electrons through the crystal.

Semiconductors-Energy Conversion

Electrical energy to light energy – Electroluminescence

Light energy to electrical energy – Photovoltaic process

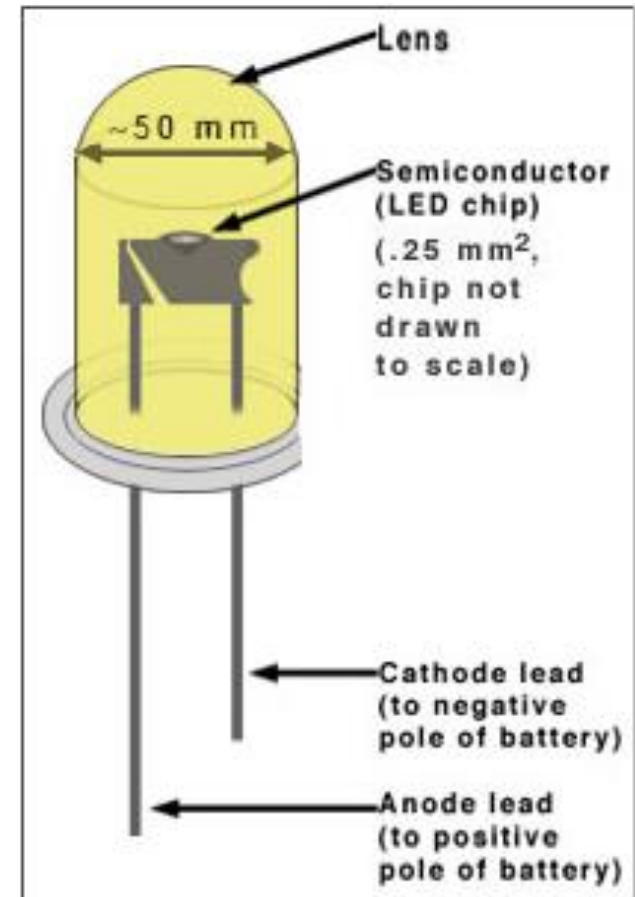
Electrical energy to light energy -
Electroluminescence

e.g. Light emitting diodes (LED)

Light Emitting Diode (LED)

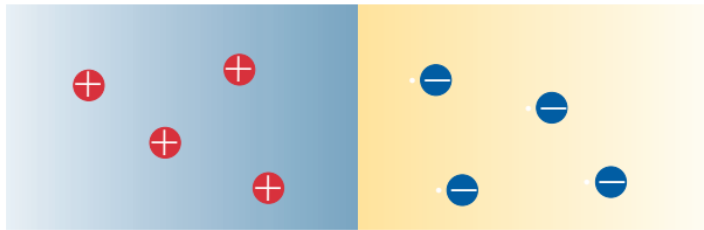
Light emitting diode is a solid state semiconductor device containing the *p-n junction*.

It converts electrical energy into light energy.

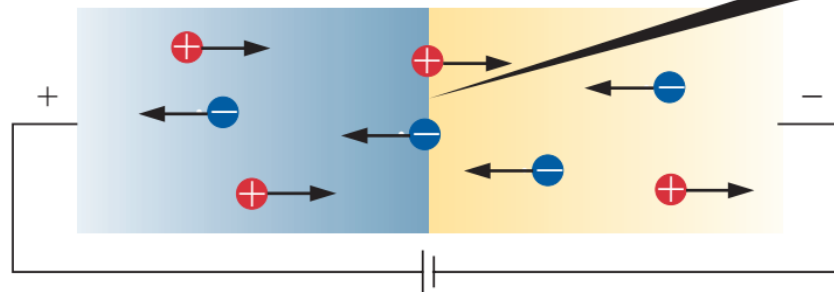


p-n junction

p n

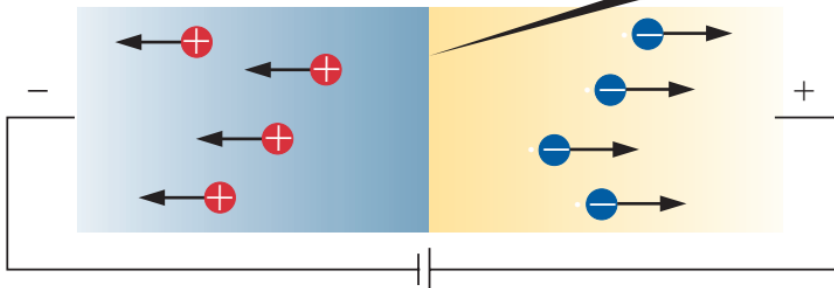


(a)



(b)

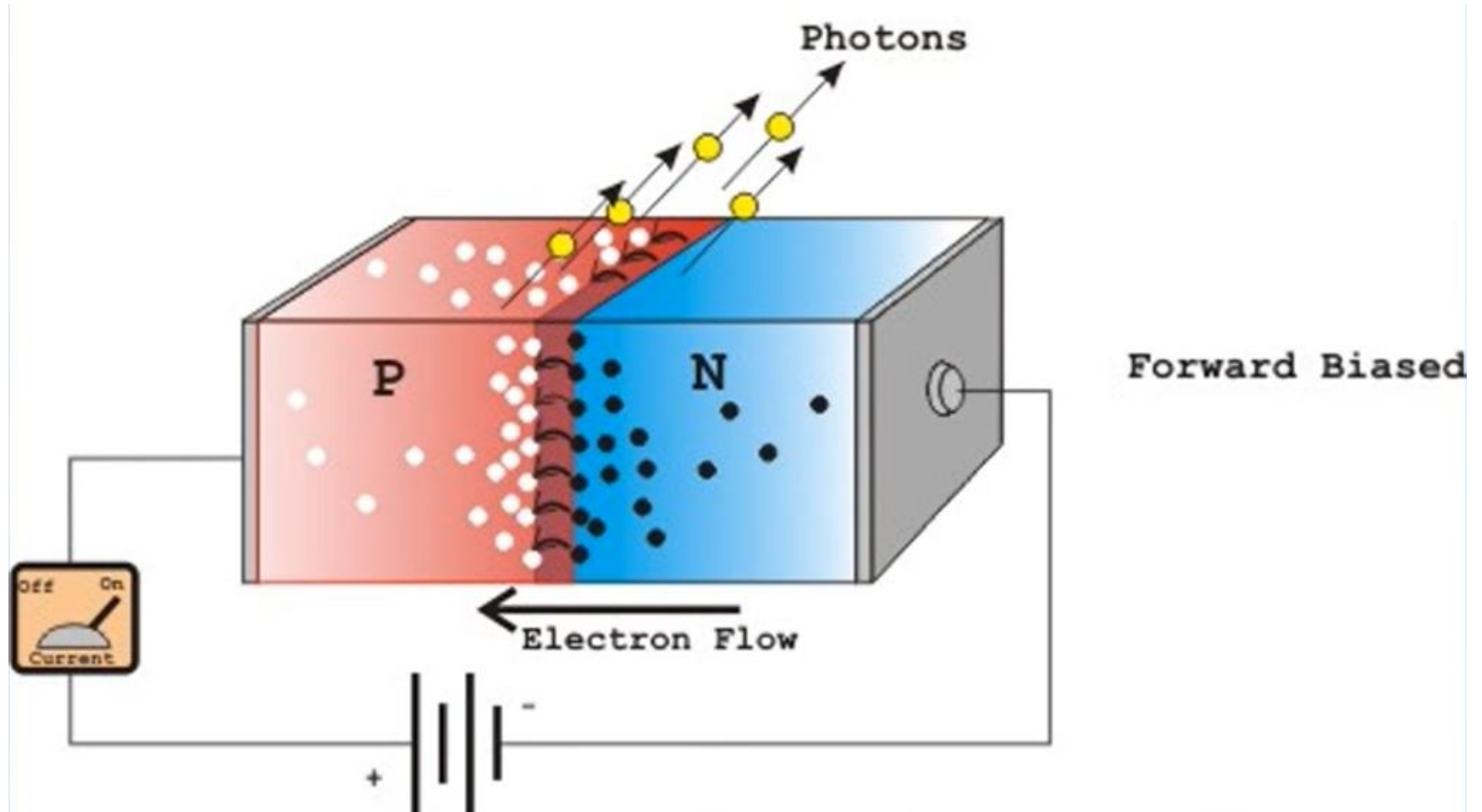
Applied voltage drives electrons and holes across junction, and current flows.



(c)

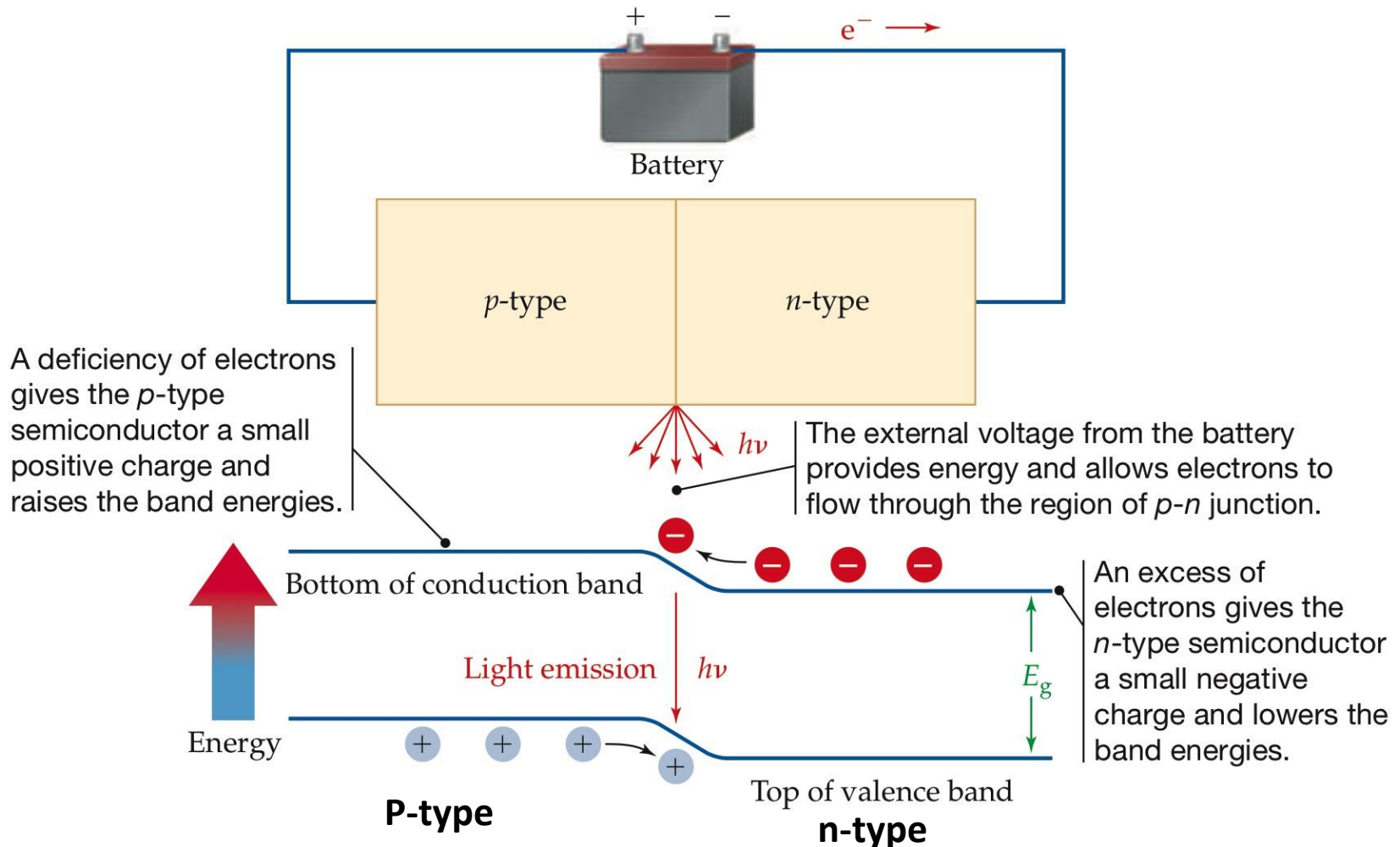
With polarity of applied voltage reversed, no current flows across junction.

How do LED emit light?

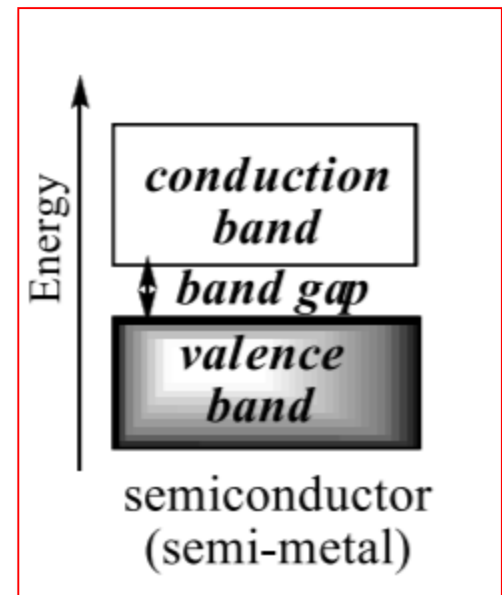
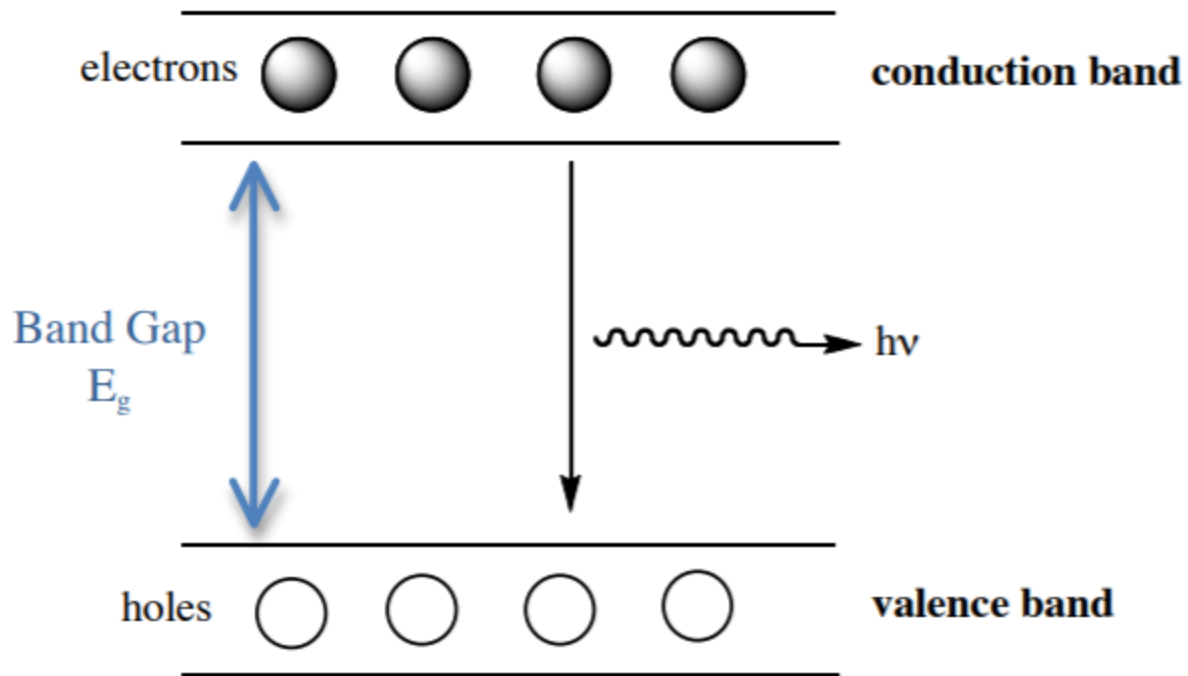


How do LED emit light?

Energy is released as light when electrons and holes combine in the p - n junction of a diode under a forward bias because of an **electronic transition** from the conduction band to the valence band



What is the origin of **different colors** in LEDs?



Band gap and emission color

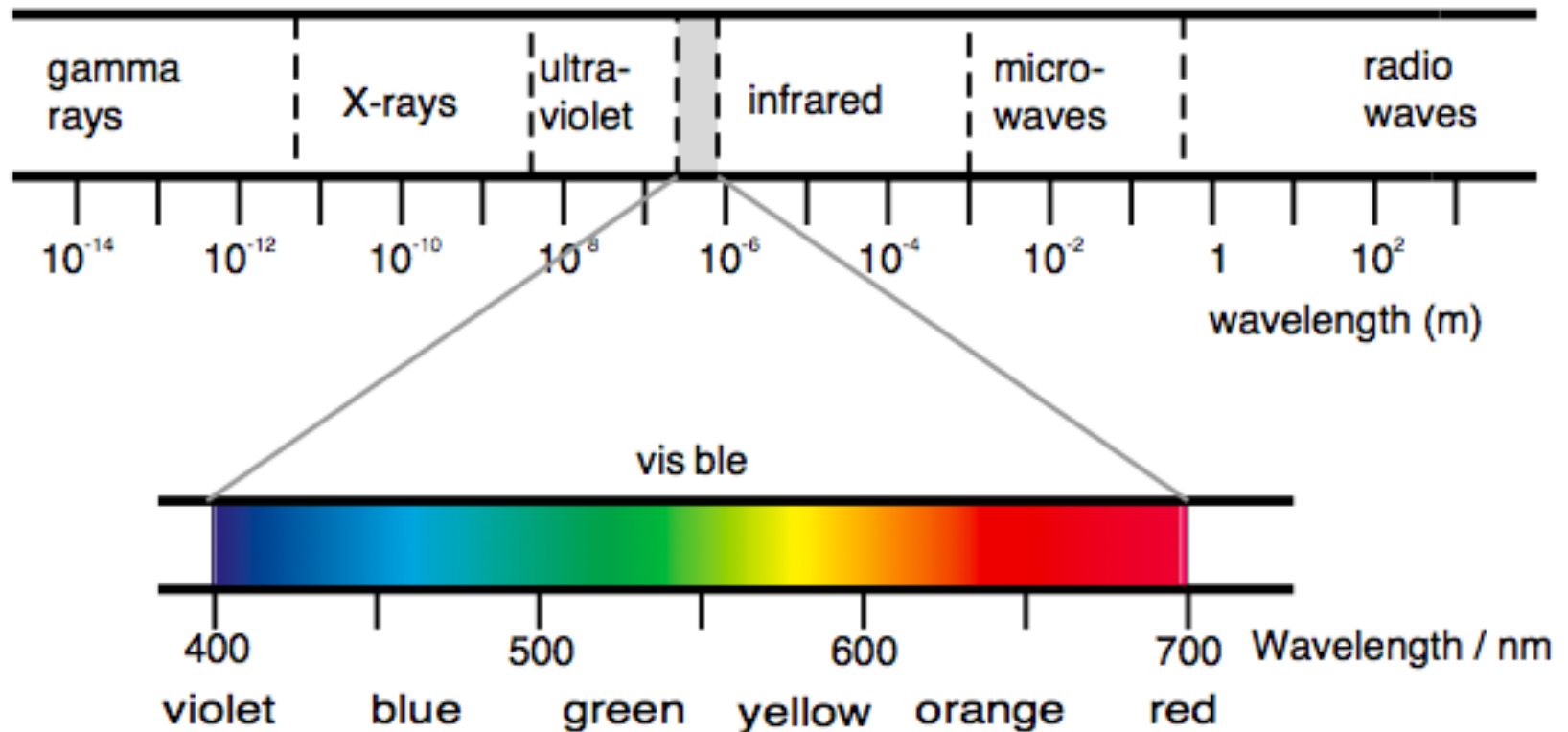
$$E = h\nu = hc/\lambda;$$

Table 1: Relationship between LED Color, Wavelength and Energy of Light.

Color of Light	Wavelength (nm)	Energy (eV & kJ/mol)
Violet	410	3.0 (290)
Blue	480	2.6 (250)
Green	530	2.3 (225)
Yellow	580	2.1 (205)
Orange	610	2.0 (195)
Red	680	1.8 (175)

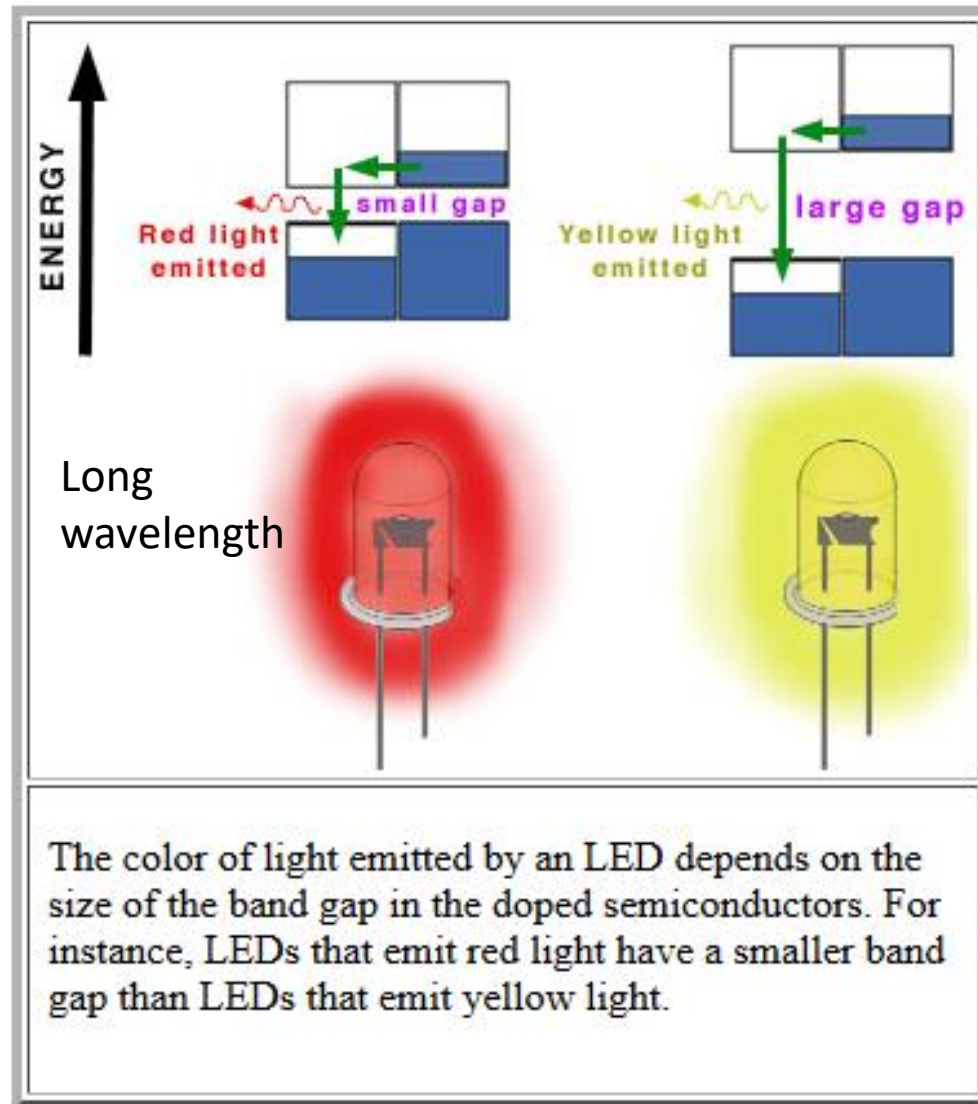
Visible Spectrum-LED Emission

The visible light emitted by the LED falls in the region of 400-700 nm of electromagnetic radiation.



What determines the colour of the emitted light?

The colour of the emitted radiation is determined by the *band gap energy* E_g



$$E = h \nu$$

$$E = h c / \lambda$$

Short wavelength

How to change the **band gap** of a material?

The band gap of a binary semiconductor can be altered *by adding an additional dopant*.

For **GaAs**, E_g is 138 kJ mol^{-1} .

By replacing **x** amount of **As** with **P** leads to form the ternary mixture.
The ternary system **GaAs_{1-x}P_x**

Depending on the amount (x) of **P**, the **dopant**, the band gap changes and hence the colour.

Amount of P	GaAs _{1-x} P _x x=0	GaAs _{1-x} P _x x=0.4	GaAs _{1-x} P _x X=1
Band gap E_g	138 kJ mol^{-1}	184 kJ mol^{-1}	218 kJ mol^{-1}
Emitted light	Infrared	Visible light Red	Visible light Green
Wavelength	870 nm	650 nm	550 nm

Calculation of Wave Length of an Emitted Radiation

$$E = h\nu \quad (\text{Planck's formula})$$

h = Planck's constant ,

E = Band gap corresponding to the emission = E_g

ν = Frequency of the emitted radiation

$$\nu = c/\lambda$$

c = Speed of light in vacuum

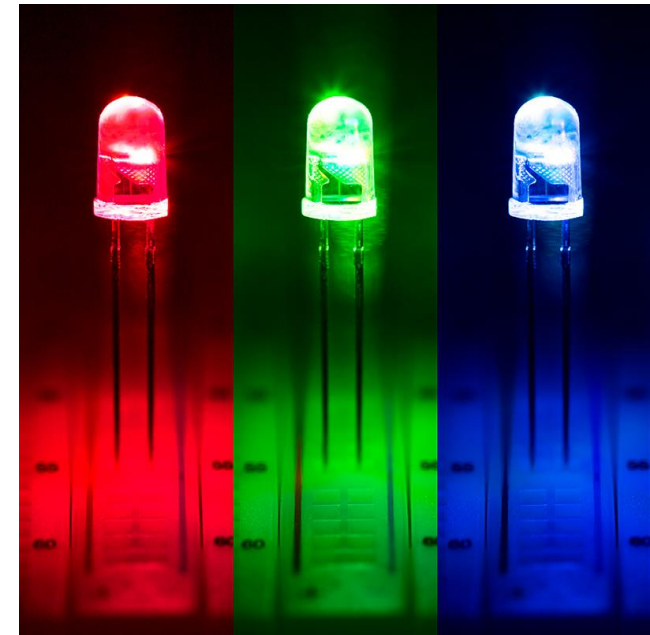
λ = Wavelength of the emitted radiation

For a given band gap energy E_g , the wave length of the radiation can be calculated

$$E = hc/\lambda ; \quad \lambda = hc/E$$

Example of various LEDs

- Gallium Arsenide (GaAs) – infra-red
- Gallium Arsenide Phosphide (GaAsP) – red to infra-red, orange
- Aluminium Gallium Arsenide Phosphide (AlGaAsP) – high-brightness red, orange-red, orange, and yellow
- Gallium Phosphide (GaP) – red, yellow and green
- Aluminium Gallium Phosphide (AlGaP) – green
- Gallium Nitride (GaN) – green, emerald green
- Gallium Indium Nitride (GaInN) – near ultraviolet, bluish-green and blue
- Silicon Carbide (SiC) – blue as a substrate
- Zinc Selenide (ZnSe) – blue
- Aluminium Gallium Nitride (AlGaN) – ultraviolet



What is the λ_{max} of a radiation emitted from LED semiconductor that has a band gap energy of 200 kJ mol^{-1} ?

Use the Planck's formula to calculate the λ_{max}

$$\lambda_{\text{max}} = hc/E$$

$$h c = 6.626 \times 10^{-34} \text{ J s} \times 3.0 \times 10^8 \text{ m s}^{-1} = 19.87 \times 10^{-26} \text{ J m}$$

$$\lambda_{\text{max}} = \frac{19.87 \times 10^{-26} \text{ J m}}{200 \text{ kJ mol}^{-1}}$$

In order to cancel the J in numerator, we need to express the energy in **joule** (J) in denominator

$$\frac{200 \text{ kJ mol}^{-1}}{6.023 \times 10^{23} \text{ mol}^{-1}} \times \frac{1000 \text{ J}}{1 \text{ kJ}} = \frac{200000 \text{ J}}{6.023 \times 10^{23}} = 3.32 \times 10^{-19} \text{ J}$$

$$\lambda_{\text{max}} = \frac{19.87 \times 10^{-26} \text{ J m}}{3.32 \times 10^{-19} \text{ J}} = 5.98 \times 10^{-7} \text{ m} = 598 \text{ nm}$$

Practice problem

Find out emission wave length for these LED semiconductors and find out which part of the electromagnetic spectrum they belong.

	GaAs	InSb	GaP	InAs	InP	AlN	AlP
Bang Gap kJ mol ⁻¹	138	17	218	34	130	411	236

Problem:

It is apparent from the table that the light emitted from these semiconductor LED materials falls in wide spectral window of electromagnetic radiation. Arrange them in increasing order of band gap.

Amount (x) of P doped in GaAs	$\text{GaAs}_{1-x}\text{P}_x$ X=0	$\text{GaAs}_{1-x}\text{P}_x$ X=1	$\text{GaAs}_{1-x}\text{P}_x$ X=0.4
Emitted light	Infrared	Visible light (Green)	Visible light (Red)

Problem:

Let's take examples of three LEDs: blue, green and red. Blue LED is made of Gallium Indium Nitride (GaInN), green LED is made of Aluminium Gallium Phosphide (AlGaP), and red LED is made of Aluminium Gallium Arsenide Phosphide (AlGaAsP). What can you comment on the band gap of GaInN, AlGaP and AlGaAsP?

If **Ge** crystals has been doped with arsenic, it will behave as n -or p -type semiconductor? Demonstrate the change with band theory.

Periodic Table of the Elements																		VIIIA 8A	
1 H Hydrogen 1.008													13 B Boron 10.811	14 C Carbon 12.011	15 N Nitrogen 14.007	16 O Oxygen 15.999	17 F Fluorine 18.998	18 Ne Neon 20.180	
3 Li Lithium 6.941	4 Be Beryllium 9.012													5 Al Aluminum 26.982	6 Si Silicon 28.086	7 P Phosphorus 30.974	8 S Sulfur 32.066	9 Cl Chlorine 35.453	10 Ar Argon 39.948
11 Na Sodium 22.99	12 Mg Magnesium 24.305	3 IIIB 3B	4 IVB 4B	5 VB 5B	6 VIB 6B	7 VIIB 7B	8 VIII 8	9 VIII 8	10 VIII 8	11 IB 1B	12 IIB 2B								
19 K Potassium 39.098	20 Ca Calcium 40.078	21 Sc Scandium 44.956	22 Ti Titanium 47.867	23 V Vanadium 50.942	24 Cr Chromium 51.996	25 Mn Manganese 54.938	26 Fe Iron 55.845	27 Co Cobalt 58.933	28 Ni Nickel 58.693	29 Cu Copper 63.546	30 Zn Zinc 65.38	31 Ga Gallium 69.723	32 Ge Germanium 72.631	33 As Arsenic 74.922	34 Se Selenium 78.971	35 Br Bromine 79.904	36 Kr Krypton 83.789		
37 Rb Rubidium 85.468	38 Sr Strontium 87.62	39 Y Yttrium 88.906	40 Zr Zirconium 91.224	41 Nb Niobium 92.906	42 Mo Molybdenum 95.95	43 Tc Technetium 98.907	44 Ru Ruthenium 101.07	45 Rh Rhodium 102.906	46 Pd Palladium 106.42	47 Ag Silver 107.868	48 Cd Cadmium 112.414	49 In Indium 114.818	50 Sn Tin 118.711	51 Sb Antimony 121.760	52 Te Tellurium 127.6	53 I Iodine 126.904	54 Xe Xenon 131.294		
55 Cs Cesium 132.905	56 Ba Barium 137.328	57-71 Lanthanide Series	72 Hf Hafnium 178.49	73 Ta Tantalum 180.948	74 W Tungsten 183.84	75 Re Rhenium 186.207	76 Os Osmium 190.23	77 Ir Iridium 192.217	78 Pt Platinum 195.085	79 Au Gold 196.967	80 Hg Mercury 200.592	81 Tl Thallium 204.383	82 Pb Lead 207.2	83 Bi Bismuth 208.980	84 Po Polonium [209]	85 At Astatine 209	86 Rn Radon 222		
87 Fr Francium 223	88 Ra Radium 226	89-103 Actinide Series	104 Rf Rutherfordium [261]	105 Db Dubnium [262]	106 Sg Seaborgium [266]	107 Bh Bohrium [264]	108 Hs Hassium [269]	109 Mt Meitnerium [278]	110 Ds Darmstadtium [281]	111 Rg Roentgenium [280]	112 Cn Copernicium [285]	113 Nh Nihonium [286]	114 Fl Flerovium [289]	115 Mc Moscovium [288]	116 Lv Livermorium [293]	117 Ts Tennessine [294]	118 Og Oganesson [294]		
			57 La Lanthanum 138.905	58 Ce Cerium 140.116	59 Pr Praseodymium 140.908	60 Nd Neodymium 144.243	61 Pm Promethium 144.913	62 Sm Samarium 150.36	63 Eu Europium 151.964	64 Gd Gadolinium 157.25	65 Tb Terbium 158.925	66 Dy Dysprosium 162.500	67 Ho Holmium 164.930	68 Er Erbium 167.259	69 Tm Thulium 168.934	70 Yb Ytterbium 173.055	71 Lu Lutetium 174.967		
			89 Ac Actinium 227.028	90 Th Thorium 232.038	91 Pa Protactinium 231.036	92 U Uranium 238.029	93 Np Neptunium 237.048	94 Pu Plutonium 244.064	95 Am Americium 243.061	96 Cm Curium 247.070	97 Bk Berkelium 247.070	98 Cf Californium 251.080	99 Es Einsteinium [254]	100 Fm Fermium 257.095	101 Md Mendelevium 258.1	102 No Nobelium 259.101	103 Lr 		

Conversion of kJ mol^{-1} unit to eV

To convert from **kJ mol^{-1}** to eV atom^{-1} divide by 96.485

Example, Converting 200 kJ mol^{-1} to eV

$$200 \text{ kJ mol}^{-1} / 96.485 = 2.07 \text{ eV}$$

Light energy to electrical energy –
Photovoltaic process

Example : Solar Cells



● **Solar Cells** One of the more common types of solar cell is, in essence, a p-n junction with an n-type surface exposed to a light source. The light provides the energy to free electrons from the n-type region and accelerate them through an external circuit into the p-type region, thus producing a current to power a calculator, light a bulb, and so on. Arrays of solar cells supply electric power to many residences (*see photo*) and businesses, as well as to space shuttles and most communications satellites.

